Astro 142 Discussion 1

2025-09-30

Dyson Lewis

Short Answer Questions

Describe the difference between storage and memory

Memory is short term access, and volatile. It stores information that the processor is currently using (that’s not in cache).

Storage is long term, and stable. It stores information that needs to be saved, and not accessed quickly.

What is assembly code?

“Readable” machine code. It’s code that doesn’t need to be compiled, instead it’s assembled (translated) into machine code.

Describe some differences between GPUs and CPUs

CPU: Executes sequentially, multiple cores can run in parallel, but each one is sequential.. Few cores, ~4-16.

GPU: Designed to computes numbers using parallel processing with ~2,000-10,000 cores. Plugs into the motherboard, and the CPU tells it what to compute using the PCIE bus.

Discussion Activity

Discussion 1 Activity

Write a python script that imports matplotlib, numpy, and scipy

Have the script print “hello world” when run

Run your script from the terminal. Paste a screenshot of your source code and the output on the next slide

When you’re done, download these slides as a pdf and upload them to the discussion 1 assignment on BruinLearn, then start Homework 1

Screenshot

```
dyson@dpl-linux: ~/ASTR142/week1
dyson@dpl-linux:~/ASTR142/week1$ nano disc1act.py
dyson@dpl-linux:~/ASTR142/week1$ cat disc1act.py
import matplotlib
import numpy
import scipy

print("Hello, world!")

#ASTR142 is a good class
dyson@dpl-linux:~/ASTR142/week1$ python disc1act.py
Hello, world!
dyson@dpl-linux:~/ASTR142/week1$ 
```